

P1: DROUGHT SEVERITY MAPPING USING VEGETATION CONDITION INDEX (VCI)

**Project Report Submitted in fulfillment of the Requirements
for the Award of the Internship of Summer Training Program
Space Science Technology**

Enrolled Subject : Remote Sensing & GIS Using Python

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1. Title

Drought Severity Assessment using Multi-temporal Vegetation Condition Index (VCI) from Satellite Data

2. Objective

The objective of this project is to assess drought severity in the Bundelkhand region using multi-temporal satellite-derived vegetation data. The Vegetation Condition Index (VCI) was calculated from MODIS NDVI datasets to identify vegetation stress and drought-affected areas. The study aims to classify drought severity into different categories and generate a thematic drought map for effective environmental monitoring and decision-making.

3. Study Area

Study Area

Bundelkhand Region, India

The study area covers selected districts of the Bundelkhand region located in Uttar Pradesh and Madhya Pradesh. The region is well known for recurrent drought conditions, irregular rainfall patterns, and agricultural vulnerability , making it suitable for drought assessment studies.

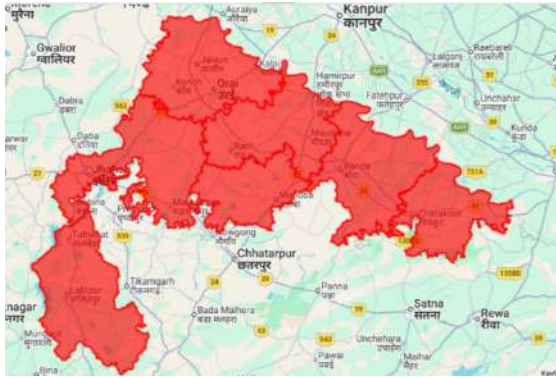


Figure 1. Area of Interest (AOI): Bundelkhand Region

4. Data Used:

Satellite Data:

<i>Parameter</i>	<i>Details</i>
<i>Satellite</i>	<i>MODIS</i>
<i>Product</i>	<i>MOD13Q1</i>
<i>Variable</i>	<i>NDVI</i>
<i>Temporal Coverage</i>	<i>2015–2025</i>
<i>Spatial Resolution</i>	<i>250 m</i>
<i>Platform</i>	<i>Google Earth Engine</i>

Software:

- *Google Earth Engine*
- *QGIS*

Data Sources:**1. MODIS MOD13Q1 NDVI Dataset:**

https://developers.google.com/earth-engine/datasets/catalog/MODIS_061_MOD13Q1

2. Google Earth Engine:

<https://earthengine.google.com/>

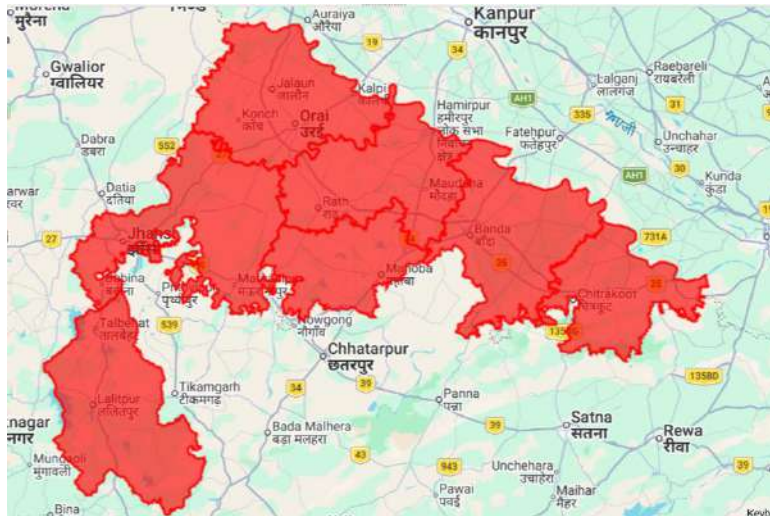
3. FAO GAUL Administrative Boundaries:

https://developers.google.com/earth-engine/datasets/catalog/FAO_GAUL_2015_level2

5. Methodology**Step 1: AOI Selection**

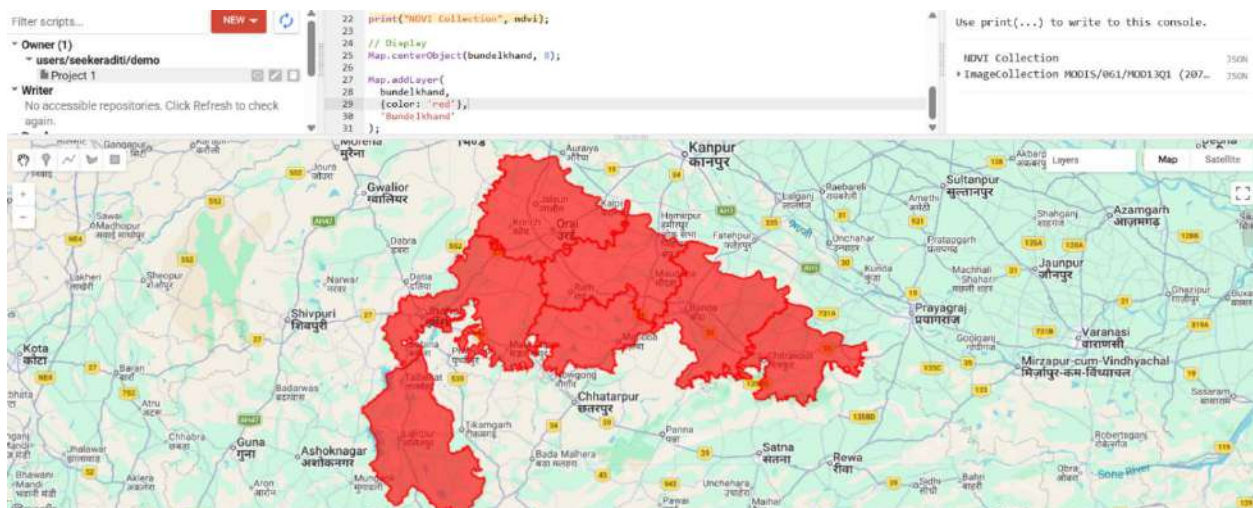
The Bundelkhand region was selected as the Area of Interest (AOI) due to its recurrent drought conditions. District boundaries were obtained from the FAO GAUL administrative boundary dataset and visualized in Google Earth Engine.

$$VCI = (NDVI - NDVI_{min}) / (NDVI_{max} - NDVI_{min})$$



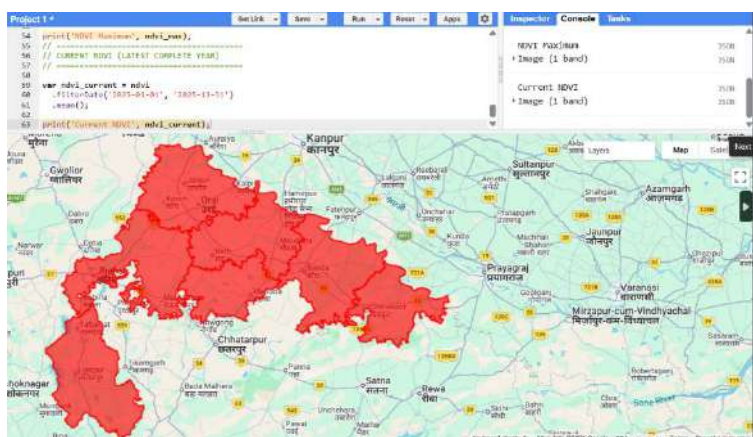
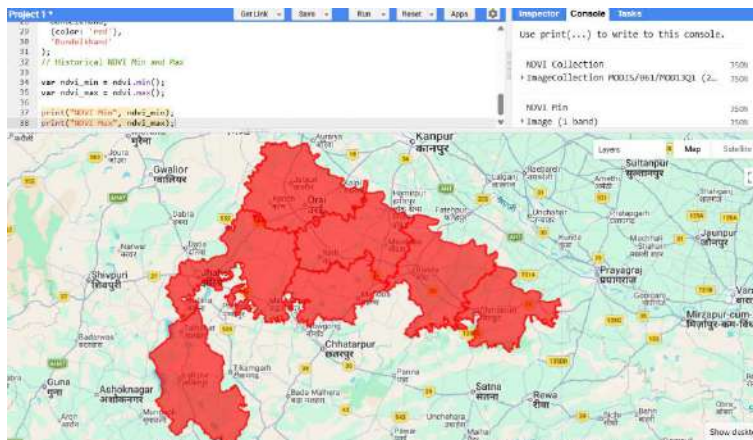
Step 2: MODIS NDVI Time-Series Loading

The MODIS NDVI (Normalized Difference Vegetation Index) time-series dataset was loaded into Google Earth Engine (GEE) to monitor vegetation conditions over the study period. The image collection was filtered based on the selected Area of Interest (AOI) to retain only the relevant satellite observations. As NDVI is a reliable indicator of vegetation health and greenness, the processed time-series data provided the foundation for analyzing temporal changes in vegetation and served as the primary input for the subsequent calculation of the Vegetation Condition Index (VCI).



Step 3: Calculation of NDVI Minimum and NDVI Maximum

The minimum (NDVI_{min}) and maximum (NDVI_{max}) values were derived from the multi-temporal MODIS NDVI dataset for the selected study period. These values represent the lowest and highest vegetation conditions recorded for each pixel and establish the reference range for vegetation health. The calculated NDVI_{min} and NDVI_{max} layers were then used as inputs for the computation of the Vegetation Condition Index (VCI).



Step 4: VCI Calculation

The Vegetation Condition Index (VCI) was derived by integrating the current NDVI values with the previously computed NDVI_{min} and NDVI_{max} layers. This process expresses the existing vegetation condition relative to its historical range, allowing areas affected by vegetation stress to be identified. The resulting VCI raster was subsequently utilized to evaluate and map drought severity throughout the Bundelkhand region.

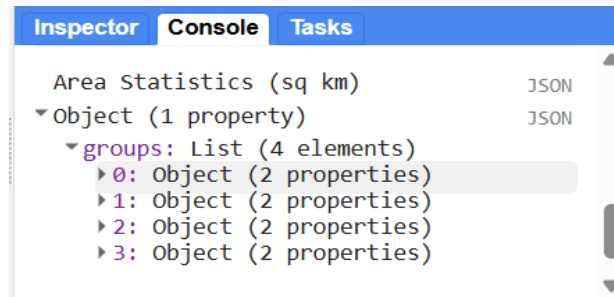
$$VCI = \frac{NDVI - NDVI_{min}}{NDVI_{max} - NDVI_{min}}$$

Step 5 : Drought Classification

Class	Category
<0.2	Extreme Drought
0.2–0.4	Moderate Drought
0.4–0.6	Mild Drought
>0.6	Healthy Vegetation

Step 6 : Area Statistics Calculation

```
Area Statistics (sq km)      JSON
▼ Object (1 property)      JSON
  ▼ groups: List (4 elements)
    ▼ 0: Object (2 properties)
      class: 1
      sum: 48.92369890674019
    ▼ 1: Object (2 properties)
      class: 2
      sum: 2485.9534645747735
    ▼ 2: Object (2 properties)
      class: 3
      sum: 30819.742593752664
    ▼ 3: Object (2 properties)
      class: 4
      sum: 3886.044000632288
```



6. Results:

VCI Map:

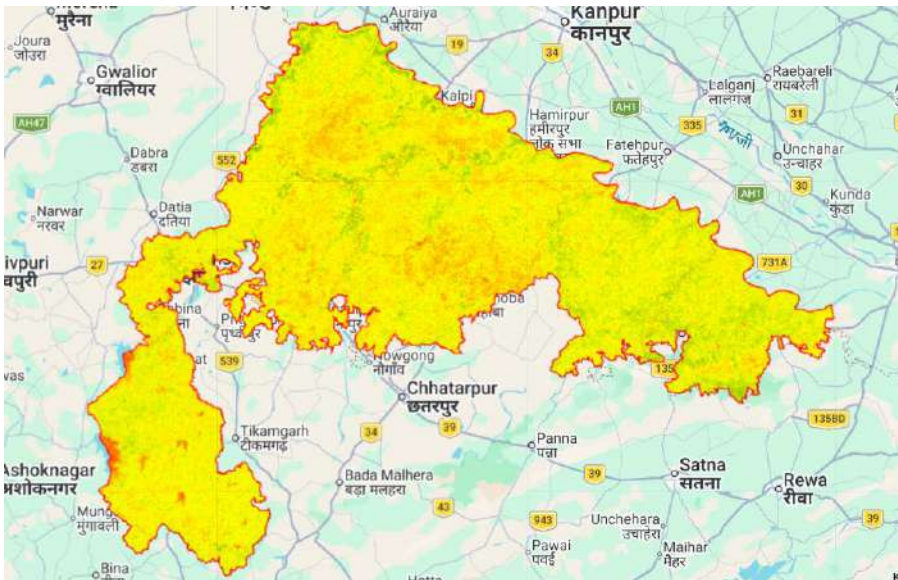


Figure 2. Vegetation Condition Index (VCI) Map

Drought Severity Map:

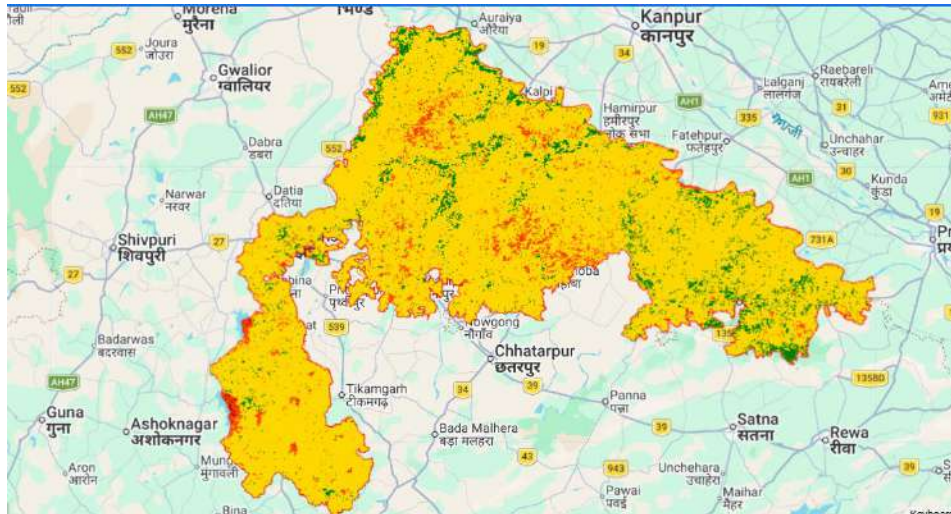


Figure 3. Drought Severity Classification Map

Area Statistics:

<i>Drought Class</i>	<i>Area (sq km)</i>
<i>Extreme Drought</i>	<i>48.92</i>
<i>Moderate Drought</i>	<i>2485.95</i>
<i>Mild Drought</i>	<i>30819.74</i>
<i>Healthy Vegetation</i>	<i>3886.04</i>

Interpretation:

The drought assessment results indicate that the majority of the Bundelkhand region falls under the Mild Drought category, covering approximately 30,819.74 sq. km. Moderate drought conditions occupy

approximately 2,485.95 sq. km, while only 48.92 sq. km is classified as Extreme Drought. Healthy vegetation conditions were observed over approximately 3,886.04 sq. km. These results suggest that although severe drought conditions are limited, vegetation stress is present across a substantial portion of the study area.

7. Conclusion:

The Vegetation Condition Index (VCI) proved to be an effective indicator for monitoring drought severity using satellite-derived vegetation information. The use of MODIS NDVI time-series data enabled the identification of vegetation stress and drought-affected regions within Bundelkhand. The study demonstrated the usefulness of remote sensing and geospatial technologies for drought assessment and environmental monitoring. Future studies may incorporate rainfall and temperature datasets to improve drought characterization and validation.

8. References:

- 1. MODIS MOD13Q1 NDVI Dataset*
- 2. Google Earth Engine Platform*
- 3. FAO GAUL Administrative Boundaries Dataset*
- 4. Remote Sensing and GIS literature related to drought monitoring*

<i>Class</i>	<i>Category</i>	<i>Area (sq km)</i>
1	<i>Extreme Drought</i>	48.92
2	<i>Moderate Drought</i>	2485.95
3	<i>Mild Drought</i>	30819.74
4	<i>Healthy Vegetation</i>	3886.04

The analysis revealed that the majority of the study area falls under the Mild Drought category, covering approximately 30,819.74 sq. km. Moderate drought conditions occupied about 2,485.95 sq. km, while only a limited area (48.92 sq. km) was identified as Extreme Drought. Healthy vegetation conditions were observed over approximately 3,886.04 sq. km.
